

**THE UNITED REPUBLIC OF TANZANIA
NATIONAL EXAMINATIONS COUNCIL OF TANZANIA
ADVANCED CERTIFICATE OF SECONDARY EDUCATION
EXAMINATION**

142/1

ADVANCED MATHEMATICS 1
(For Both School and Private Candidates)

Duration: 3 Hours

Year: 2026

Instructions

1. This paper consists of **ten (10)** questions. **wazaelimu.com**
2. Answer **all** the questions. Each question carries **ten (10)** marks.
3. All work done in answering each question must be shown clearly.
4. Non-programmable calculators and NECTA mathematical tables may be used.
5. All writing must be in **blue** or **black** ink, **except** drawing which must be in pencil.
6. You should **not** write any notes in this question paper.
7. Communication devices and any unauthorised materials are **not** allowed in the examination room.
8. Write your **Examination Number** on every page of your answer booklet(s).



1. (a) Given that $u = \log 98$, $v = \sqrt[3]{0.00651}$, $w = \sqrt{\tan 82^\circ}$, and $x = v^2 w$, use a scientific calculator to find the value of $\frac{u^2 + vw^2 \sqrt{x}}{v^2 w}$.

- (b) By using a non-programmable calculator, find the sum of mean and standard deviation for the following data, giving your answer correct to 2 significant figures.

Number	1	2	3	4	5
Frequency	18	15	38	7	2

- (c) Use a scientific calculator to evaluate $\sqrt[3]{\frac{({}^3P_2)(6!) + ({}^5C_2)(2!)}{6(4!)(5!)}}$ correct to 2 decimal places.

2. (a) Solve the equation $3\sinh x - \cosh x = 1$.

- (b) Integrate $\frac{1}{2x\sqrt{1-9x^2}}$ with respect to x .

- (c) Draw the graph of $y = \cosh^{-1} x$.

3. A carpenter produces tables and chairs. Each table yields a profit of Tshs. 3,000, while each chair yields a profit of Tshs. 1,000. The carpenter has a maximum of 40 hours per week and requires six hours to make a table and three hours to make a chair. The customer requests at least three times as many chairs as tables. Tables occupy four times the storage space of chairs, and storage limits allow a maximum of four tables per week. How many tables and chairs should be produced each week so that the carpenter earns a maximum profit?

4. The following frequency distribution table presents scores attained by students in mathematics test:

Class	0 - 9	10-19	20-29	30-39	40-49	50-59	60-69	70-79	80-89
Frequency	1	8	19	28	25	17	7	3	1

Use this information to calculate:

- (a) the mode.
 (b) the standard deviation.

5. (a) Sets A and B are such that $A = \{x \in \mathbb{R} : 2 < x \leq 5\}$ and $B = \{x \in \mathbb{R} : x \leq 2\}$. Use a number line to find $A' \cap B$.
- (b) Given non-empty sets A and B , use laws of algebra of sets to simplify the expression $(A \cup B)' \cap (A' \cap B)$.
- (c) In a class of 24 pupils, 13 study History but not English, 6 study English but not History and 3 study neither English nor History. How many pupils study History?
6. (a) Given $f(x) = 3x - 2$ and $g(x) = x^2$, determine $g^{-1}(f(x))$.
- (b) A function is defined on a domain by $f(x) = \frac{x^2}{x-2}$. Sketch the graph of $|f(x)|$.
7. (a) Using Newton-Raphson method and $x_0 = 0.75$, perform two iterations to approximate the root of the equation $e^x = 3 - x$ correct to four decimal places.
- (b) Using five ordinates with four significant figures each, use the Simpson's rule to approximate the value of $\int_0^1 \cos \sqrt{x} dx$.
8. (a) A point $P(x, y)$ moves in such a way that its distance from x -axis is always $\frac{1}{5}$ of its distance from the origin. Find the equation of its path.
- (b) A circle whose centre is on the line $x + 3y + 7 = 0$ passes through the points $A(-1, 1)$ and $B(6, 0)$. Find its equation.
9. (a) Verify that $\int x \ln(x^2 + 1) dx = \frac{1}{2} [(x^2 + 1) \ln(x^2 + 1) - (x^2 + 1)] + C$.
- (b) Calculate the volume of the solid generated by revolving the region bounded by the hyperbola $\frac{x^2}{4} - \frac{y^2}{1} = 1$ and the lines $y = 0$ and $y = 1$ about the x -axis.
10. (a) Differentiate $f(x) = \frac{1}{\sqrt{x}}$ from the first principles.
- (b) Differentiate $f(x) = \frac{x}{\sqrt{x-1}}$ with respect to x .

- (c) In a hot, dry summer, water is evaporating from a desert oasis. The volume of water remaining after t days is given by the equation $V(t) = 2(50 - t)^2 \text{ m}^3$. Find the rate at which the water is evaporating at the fifth day.